Exploratory Data Analysis

AI/ML Internship Task 1

Student Performance Data Explorer and complete Iris EDA challenge. This report presents results computed from the included UCI datasets, with data-quality counts, descriptive statistics, visualizations, and limitations.

Project contents: reusable Python analyses, two Jupyter notebooks, raw and cleaned data, nine chart files, editable Mermaid workflow diagrams, and this report.

1. Dataset descriptions and sources

The UCI Student Performance mathematics dataset (dataset 320; student-mat.csv) describes secondary students from two Portuguese schools. It contains student, family, school, and grade attributes. Its original CSV uses semicolon delimiters. G3 is presented as Math Score; G1 and G2 are prior period grades.

The UCI Iris dataset (dataset 53), credited to R. A. Fisher (1936), contains 150 flowers, three species, and four continuous measurements in centimeters. The source is licensed CC BY 4.0.

2. Data quality and cleaning

Student raw shape: 395 rows × 33 columns. Missing cells: 0; exact duplicate rows: 0. Removed duplicates: 0; imputed missing cells: 0; cleaned rows: 395; missing cells remaining: 0. The original CSV remains unchanged, and the cleaned version is separate.

Iris raw shape: 150 × 5. Missing cells: 0; duplicate rows: 3. Removed duplicates: 3; imputed cells: 0; cleaned rows: 147.

3. Student performance: descriptive results

Final mathematics grade summary (0–20): mean 10.42; median 11.00; mode 10; minimum 0; maximum 20; standard deviation 4.58. The IQR rule flags 0 potential outliers; these are not necessarily errors.

Correlation between Period 1 and final grade: 0.801; Period 2 and final: 0.905. This association is expected to be strong for sequential grades, but it does not show causation.

School	Mean final grade
GP	10.49
MS	9.85

4. Student performance visualizations

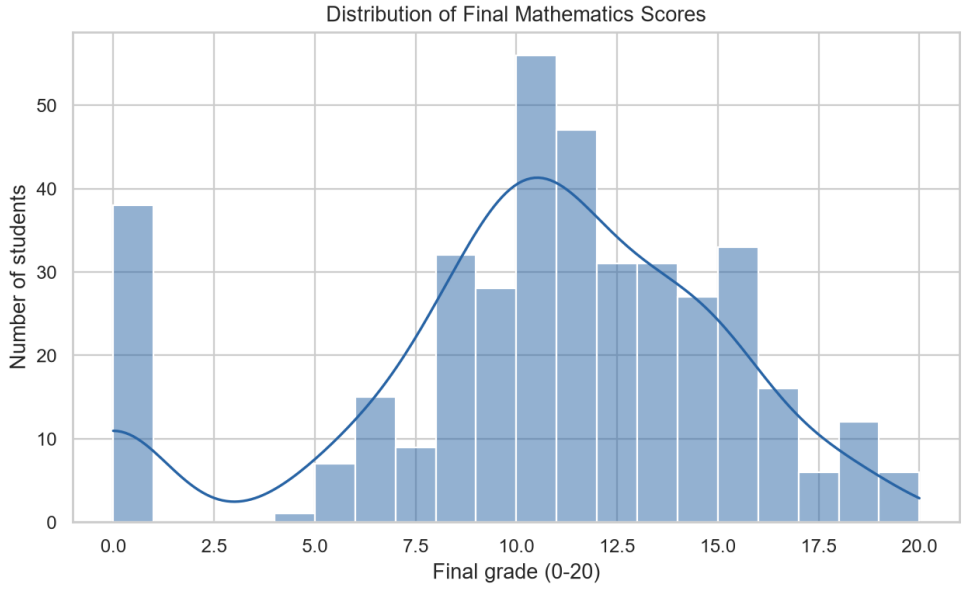


Figure 1. Distribution of final mathematics scores.

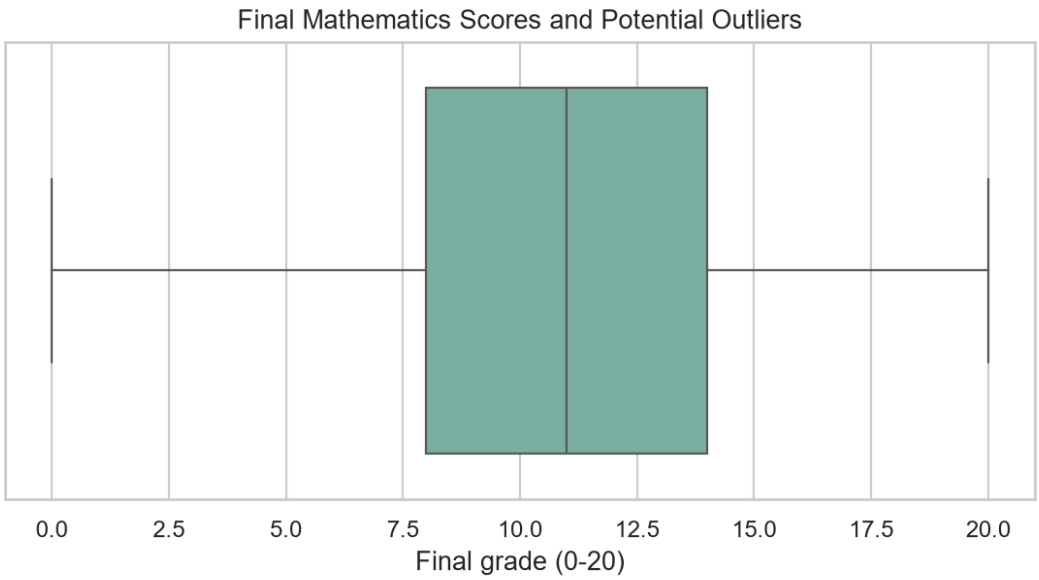


Figure 2. Final grade box plot and potential outliers.

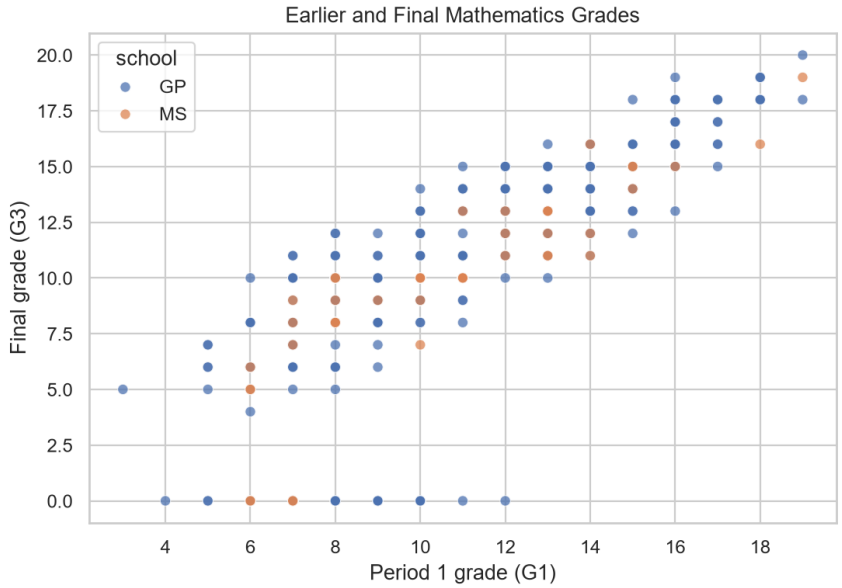


Figure 3. Earlier and final mathematics grades colored by school.

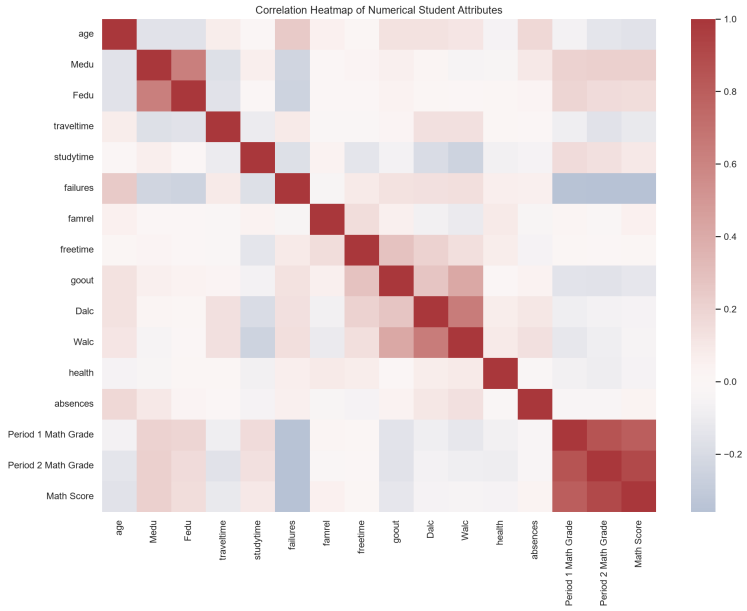


Figure 4. Correlations among numerical student attributes.

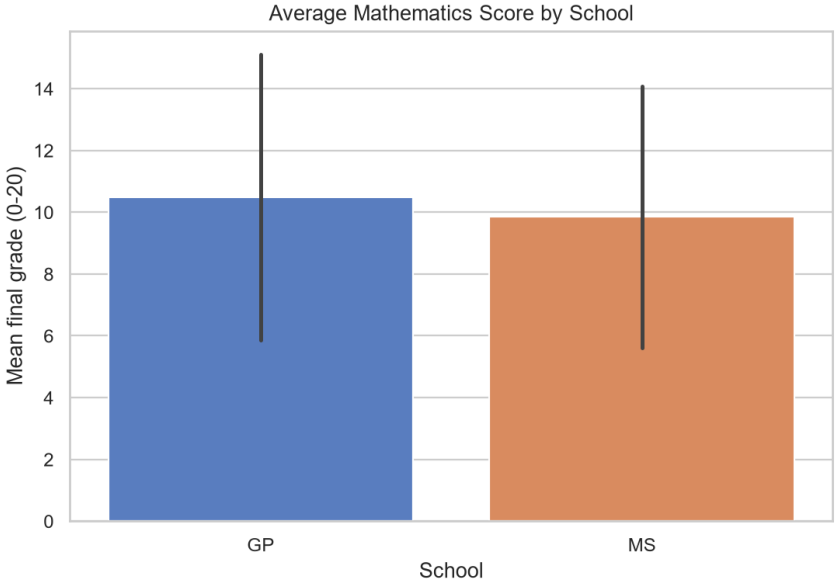


Figure 5. Mean mathematics score by school.

5. Iris challenge results

The cleaned Iris dataset contains 147 records. Species counts are Iris-setosa: 48, Iris-versicolor: 50, Iris-virginica: 49. Petal length and width differ noticeably among species; setosa forms a distinct cluster, while versicolor and virginica overlap. This is a descriptive pattern from a small classic dataset.

Species	Sepal L	Sepal W	Petal L	Petal W
Iris-setosa	5.01	3.43	1.46	0.25
Iris-versicolor	5.94	2.77	4.26	1.33
Iris-virginica	6.60	2.98	5.56	2.03

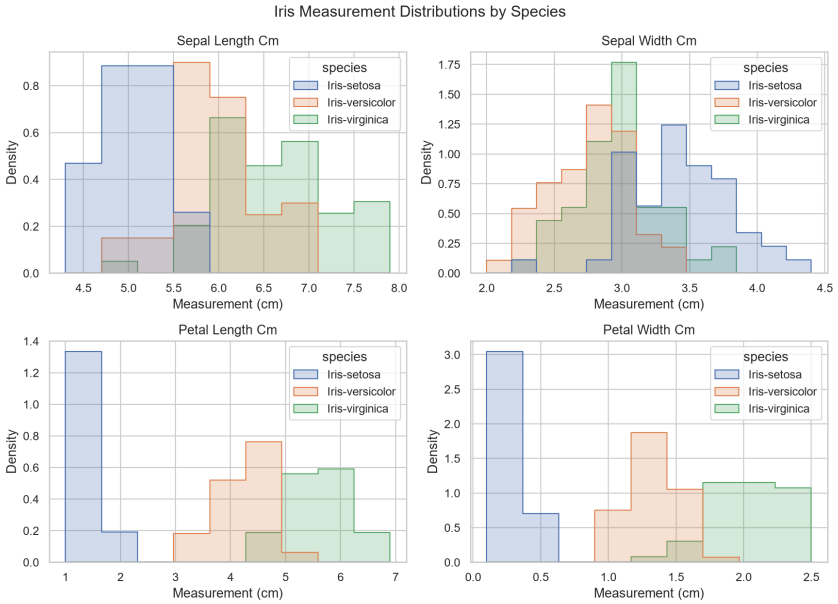


Figure 6. Histograms of four measurements by species.

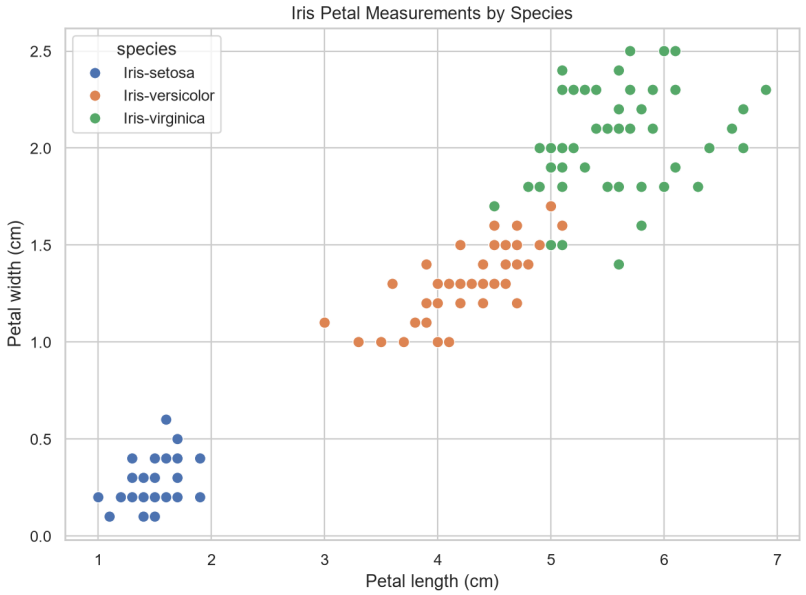


Figure 7. Species-colored scatter of petal measurements.

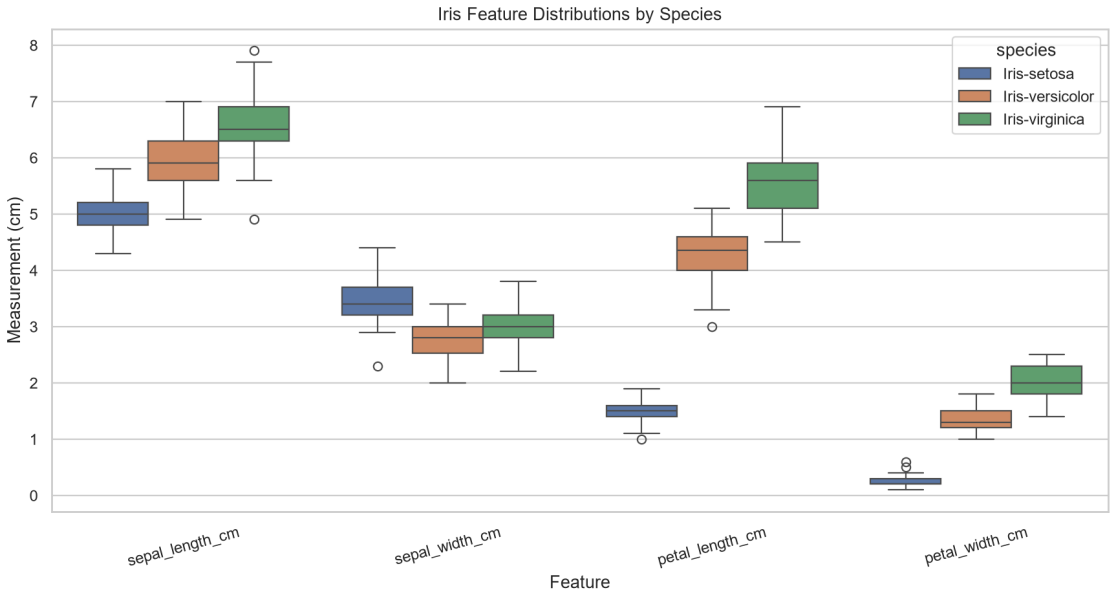


Figure 8. Measurement box plots by species.

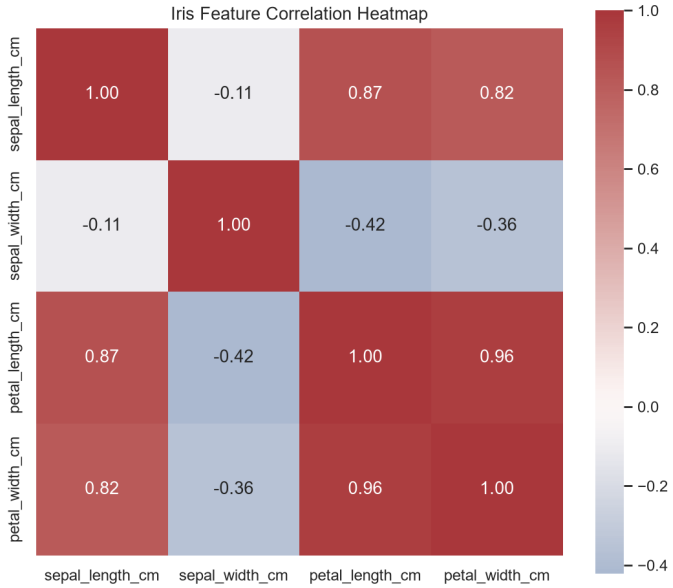


Figure 9. Correlation heatmap of Iris measurements.

6. Interpretation and limitations

Correlation describes linear co-variation and does not establish causation. Student data is observational, from two schools, and includes prior grades that are naturally related to final grades. School means do not control for confounders. The IQR rule is a screening heuristic. Iris has only 150 records and three species; its patterns should not be generalized to every population.

7. Conclusion and references

The project preserves raw data, separates cleaned outputs, includes two executable notebooks, and generates nine figures from genuine source data.

Sources: Cortez & Silva (2008), UCI Student Performance dataset 320, archive.ics.uci.edu/dataset/320/student. Fisher (1936), UCI Iris dataset 53, DOI 10.24432/C56C76 (CC BY 4.0). Tools: Python, pandas, Matplotlib, Seaborn, ReportLab.